Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics

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- Fiducial Phase Space and MC Simulations
- Event selection & Background estimation
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Experimental Setup & Target

- **Dataset:** $140\,\text{fb}^{-1}$ of pp collisions at $\sqrt{s} = 13$ TeV (2015–2018).
- **Target:** Fully leptonic decays of $W^{\pm}Zjj$ production.

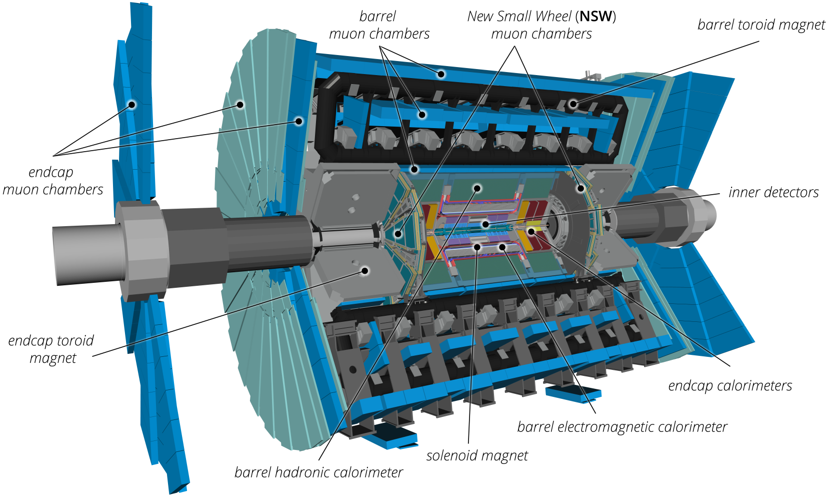
Analysis Objectives

- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).

New Physics Search

- **BSM:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).

The ATLAS Detector
1 Introduction



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Gauge Boson Self-Interactions

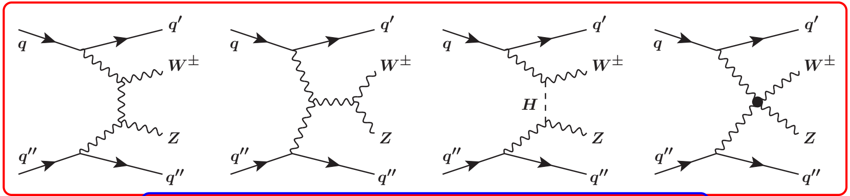
- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons ($W^{\pm}, Z$) **self-interact**.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^{\mu} W_c^{\nu} (\partial_{\mu} W_{a\nu} - \partial_{\nu} W_{a\mu}) \qquad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^{\mu} W_c^{\nu} W_{r\mu} W_{s\nu}$$

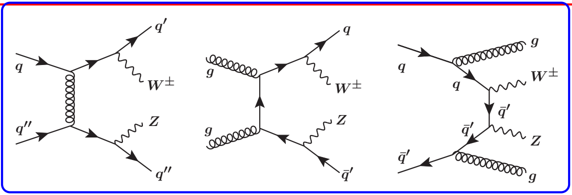
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from **New Physics**.

Experimental Signature: Vector Boson Scattering

Vector Boson Scattering (VBS) $VV \to VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^\pm Zjj$.



$W^\pm Zjj$**-EW**



$W^\pm Zjj$**-QCD**

A little look at the Effective Field Theory (EFT)

The $W^{\pm}Zjj$-EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^{\pm}Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an **EFT framework**. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective **Wilson coefficients** as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ: is the energy scale of the new processes

A little look at the Effective Field Theory (EFT)

The squared scattering amplitude of the effective field theory predicion for $W^{\pm}Zjj$ production can be written as:

$$\left| A_{\mathrm{SM}} + \sum_i c_i A_i \right|^2 = |A_{\mathrm{SM}}|^2 + \sum_i c_i \, 2\mathrm{Re}(A_{\mathrm{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j \, 2\mathrm{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)}/\Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2\,\mathrm{Re}(A_{\mathrm{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i {c_i}^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij,i \neq j} c_i c_j 2\,\mathrm{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.

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Fiducial Phase Space for Leptons

The $W^{\pm}Zjj$ cross-sections are measured in a **fiducial phase space**. Leptons produced in the decay of a hadron, a τ-lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	> 15 GeV
W lepton	Transverse momentum ($p_T^{\ell_W}$)	> 20 GeV
All leptons	Pseudorapidity ($\lvert\eta_\ell\rvert$)	< 2.5
Z boson	Invariant mass window ($\lvert m_{\ell^+\ell^-} - m_Z^{\text{PDG}}\rvert$)	< 10 GeV
W boson	Transverse mass (m_T^W)	> 30 GeV
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3

Fiducial Phase Space for Jets

In addition, at least **two jets are required**, reconstructed using the anti-k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b-quark in the initial or final states, such as tZj production, are not considered as part of the $W^{\pm}Zjj$-EW events.

Category	Kinematic Variable	Requirement
All jets	Transverse momentum (p_T^j)	> 40 GeV
All jets	Pseudorapidity ($\|\eta_j\|$)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	> 500 GeV
	Jet Veto	
b-quark jets	Absence of b-jets in initial/final state	$N_{b\text{-jet}} = 0$

Simulation and Background Overview

Signal & $W^{\pm}Zjj$ Components

Generated with `MG5_aMC@NLO` + `Pythia8`:

- **EW ($W^{\pm}Zjj-$ EW):** Signal, purely electroweak (LO, $\mathcal{O}(\alpha^6_{\text{EW}})$).
- **QCD ($W^{\pm}Zjj-$ QCD):** VBS background (NLO in QCD).
- **Interference:** EW-QCD quantum int. (LO), $+6 \div 12\%$ contribution in SR.

Other Backgrounds

- **Prompt Leptons:** ZZ, WW, VVV, $t\bar{t}V$, tZj. Estimated via MC simulations.
- **Fake Leptons:** Misidentified jets. Estimated via data-driven techniques.

Detector Simulation

All MC samples processed through `Geant4` ATLAS detector simulation and standard reconstruction software.

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Event Selection: Physics Objects Definition

Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency $\approx 99\%$ for offline-selected $W^{\pm}Zjj$ events).

Physics Objects Reconstruction:

- **Prompt Leptons** $e, \mu, \nu\ (E_T^{miss})$: Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt leptons.
- **Hadronic Jets:** Clustered using the *anti-k_t* algorithm ($R = 0.4$).
- **b-tagging:** Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).

Physics Objects: Leptons

Object	Selection Criteria
Muons	*Medium* Identification (Inner Detector + Muon Spectrometer) $p_T > 15\,\mathrm{GeV}$, $\lvert\eta\rvert < 2.5$ Impact parameters: $\lvert d_0/\sigma_{d_0}\rvert < 3.0$, $\lvert z_0 \sin\theta\rvert < 0.5\,\mathrm{mm}$
Electrons	*Medium* Likelihood Identification $p_T > 15\,\mathrm{GeV}$, $\lvert\eta\rvert < 1.37$ or $1.52 < \lvert\eta\rvert < 2.47$ Impact parameters: $\lvert d_0/\sigma_{d_0}\rvert < 5.0$, $\lvert z_0 \sin\theta\rvert < 0.5\,\mathrm{mm}$

Note: Impact parameter cuts are crucial to select ***prompt*** *leptons and reject non-prompt background from heavy-flavor decays.*

Physics Objects: Jets & $E_{\mathrm{T}}^{\mathrm{miss}}$

Object	**Selection Criteria**
Jets	Clustered via *anti-k_t* algorithm ($R = 0.4$) $p_T > 25\,\mathrm{GeV}$, $\lvert\eta\rvert < 4.5$ Pile-up suppression (JVT):*tight* ($p_T < 60\,\mathrm{GeV}$, $\lvert\eta\rvert < 2.4$) and *loose* ($p_T < 120\,\mathrm{GeV}$, $2.5 < \lvert\eta\rvert < 4.5$)
Missing Energy ($E_{\mathrm{T}}^{\mathrm{miss}}$)	Proxy for the undetected neutrino. Computed as the negative vector sum of all calibrated *hard* objects (leptons, jets) plus a track-based *soft* term.

Event Selection: Boson Reconstruction

Category	Selection Criteria
Baseline	Exactly 3 lepton candidates (μ or e) Veto on a 4th *loose* lepton ($p_T > 5\,\mathrm{GeV}$) Trigger matching: at least 1 lepton with $p_T > 25/27\,\mathrm{GeV}$
Z **Boson Candidate**	Same-Flavor Opposite-Sign lepton pair $\lvert m_{\ell\ell} - m_Z^{\mathrm{PDG}} \rvert < 10\,\mathrm{GeV}$ (If ambiguous, the pair closest to m_Z is chosen)
W **Boson Candidate**	The remaining 3rd lepton + $E_{\mathrm{T}}^{\mathrm{miss}}$ Stricter requirements: *Tight* ID & Isolation, $p_T > 20\,\mathrm{GeV}$ Transverse mass: $m_T^W > 30\,\mathrm{GeV}$

Irreducible Backgrounds

Irreducible backgrounds feature true prompt leptons mimicking the signal topology.

ZZ Control Region (ZZ-CR)

- **Background:** $ZZ \to 4\ell$ ($ZZjj$-QCD).
- **Origin:** Survives via lepton acceptance loss.
- **Selection:** Requires a 4th *loose* lepton to ensure orthogonality.

b Control Region (b-CR)

- **Background:** Top-quark ($t\bar{t} + V$, tZj).
- **Origin:** Survives via b-tagging inefficiency.
- **Selection:** Inverts the b-veto by requiring ≥ 1 b-jet.

While rare processes like **Tribosons (VVV)** and $ZZjj$**-EW** are estimated via **pure Monte Carlo** due to their low yields, the dominant sources are constrained using dedicated **Control Regions**:

Reducible Backgrounds (Fake Leptons)

Unlike irreducible processes, reducible backgrounds in the $W^{\pm}Zjj$ phase space arise from instrumental misidentification rather than true prompt leptons.

Fake E_T^{miss} & Misidentified Objects

- $Z+$ **jets:** A hadronic jet (e.g., charged pion) is misidentified as a lepton. E_T^{miss} is faked by finite Jet Energy Resolution.
- $Z\gamma$: A true photon mimics an electron.

True E_T^{miss} & Non-Prompt Leptons

- $t\bar{t}$ **&** Wt: True E_T^{miss} from $W \to \ell\nu$. The 3rd lepton is a non-prompt decay from a b-hadron inside a jet.
- WW: Two prompt leptons and true neutrinos. The 3rd lepton is faked by associated QCD jets or pile-up tracks.

Estimation Strategy: Data-Driven Matrix Method

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Signal Extraction: Multivariate Analysis (BDT)

The Challenge

$WZjj$-EW (signal, $\mathcal{O}(\alpha_{\text{EW}}^6)$) and $WZjj$-QCD (background, $\mathcal{O}(\alpha_s^2\alpha_{\text{EW}}^4)$) share the exact same final state. Kinematic cut-based separation is sub-optimal.

$\Rightarrow$ **Solution:** Boosted Decision Tree (TMVA).

BDT Input Variables (15 observables):

- *Jet:* m_{jj}, $|\Delta\eta_{jj}|$, $\Delta\phi_{jj}$, N_{jets}
- *Boson:* p_T^W, p_T^Z, m_T^{WZ}
- *Correlations:* Lepton centrality, event balance

BDT Output:

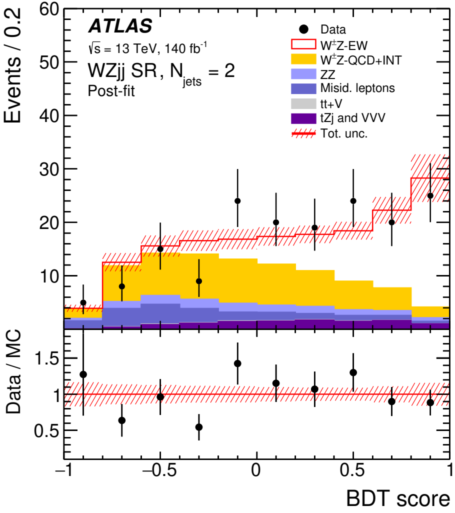
- Maps a multi-dimensional phase space into a **single discriminant score**.
- Used as the primary observable for signal extraction.

Note: An independent auxiliary BDT is deployed in the b-CR to differentiate between $t\bar{t}V$ and tZj.

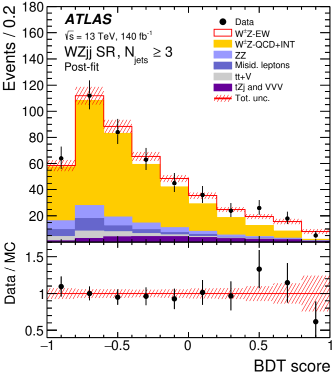
The Global Likelihood Fit & Post-fit distributions SR ($N_{jets} = 2$)

Fit Components

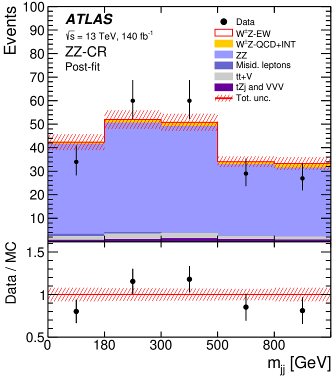
- **Regions (r):** SR ($N_{\text{jets}} = 2$), SR ($N_{\text{jets}} \geq 3$), b-CR, ZZ-CR.
- **Parameters of Interest (μ):** Strength modifiers μ_{EW} and μ_{QCD}.
- **Unconstrained Parameters (K):** Background normalizations ($K_{t\bar{t}V}$, K_{tZj}, K_{ZZ}) extracted dynamically from CRs.
- **Nuisance Parameters (ν):** Theoretical/experimental systematics, constrained by a Gaussian penalty term.



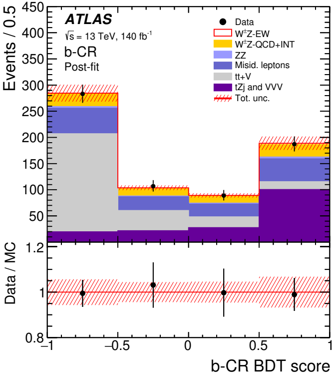
Post-fit distributions SR ($N_{jets} \geq 3$), ZZ-CR and b-CR



(a)



(b)



(c)

Fiducial Cross-Section Extraction

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors (μ_{EW}, μ_{QCD}) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-\mathrm{EW}} = \sum_i \mu_i^{WZjj-\mathrm{EW}} \cdot \sigma_{i,th.MC}^{WZjj-\mathrm{EW}}$$

$$\sigma_{WZjj-\mathrm{strong}} = \sum_i \left(\mu_i^{WZjj-\mathrm{QCD}} \cdot \sigma_{i,th.MC}^{WZjj-\mathrm{QCD}} + \mu_i^{WZjj-\mathrm{INT}} \cdot \sigma_{i,th.MC}^{WZjj-\mathrm{INT}} \right)$$

**Index i runs over the two SR categories: $N_{jets} = 2$ and $N_{jets} \geq 3$.*

- **Interference term:** μ_{INT} is constrained as $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$.

Results and Comparison with SM Predictions

VBS (EW Production)

$$\sigma_{\text{obs}}^{\text{EW}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}}\,\text{fb}$$

$$\sigma_{\text{th}}^{\text{EW}} = 0.370 \pm 0.030\,\text{fb}$$

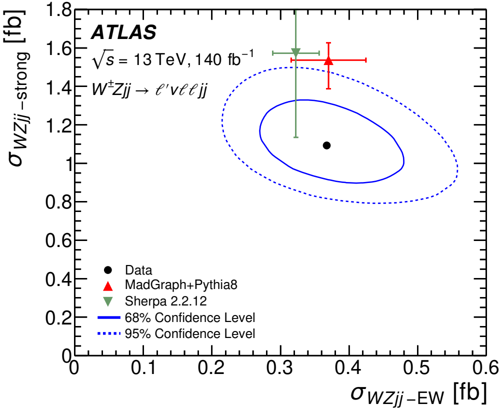
$\Rightarrow$ Excellent SM agreement

Strong + Int.

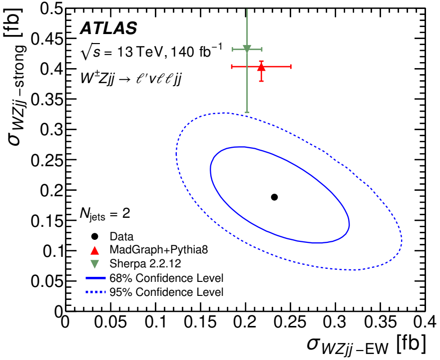
$$\sigma_{\text{obs}}^{\text{strong}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}}\,\text{fb}$$

$$\sigma_{\text{th}}^{\text{strong}} = 1.537 \pm 0.150\,\text{fb}$$

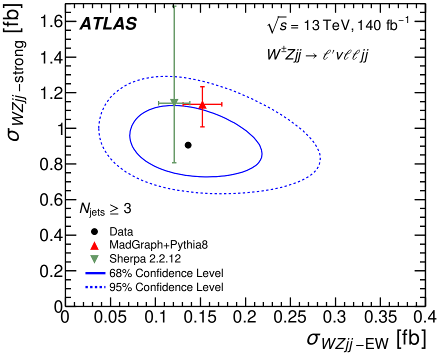
$\Rightarrow$ Deficit factor $\sim$ 0.7 (1.8σ tension)



$W^{\pm}Zjj-$ EW and $W^{\pm}Zjj$-strong ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

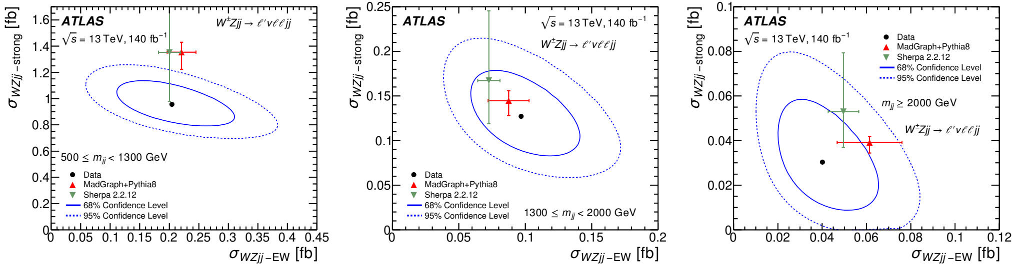


(a)



(b)

$W^{\pm}Zjj-$ EW and $W^{\pm}Zjj$-strong in differents m_{jj} bins



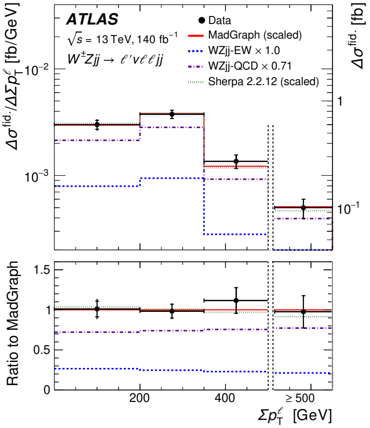
(a) (b) (c)

$W^{\pm}Zjj$ differential measurements

Differential measurements of $W^{\pm}Zjj$ production cross-section are performed as a function of variables sensitive to anomalies in the quartic gauge coupling.

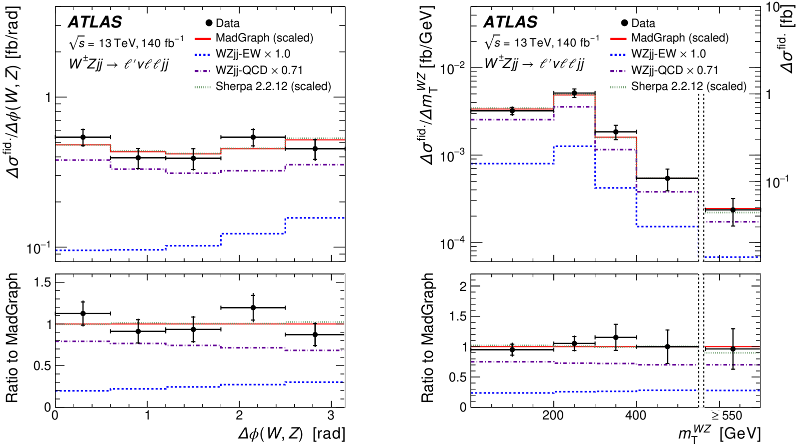
Kinematic Variables

- $\sum p_T^{\ell}$: Scalar sum of the transverse momenta of the three charged leptons (W, Z).
- $\Delta\phi(\mathbf{W}, \mathbf{Z})$: Difference in azimuthal angle between the W and Z bosons' directions.
- $\mathbf{m}_T^{\mathbf{WZ}}$: Transverse mass of the $W^{\pm}Z$ system.



$W^{\pm}Zjj$ differential measurements

5 Measurement methodology and results



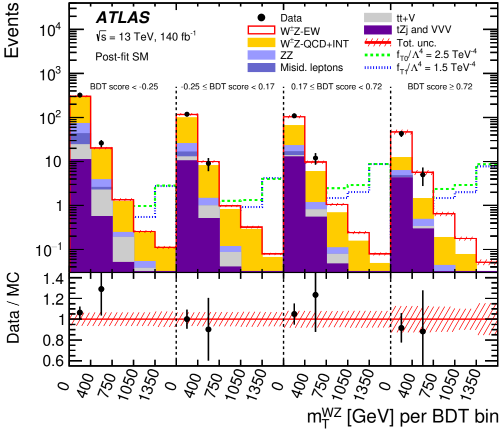
(a) (b)

Search for aQGC: Methodology

Anomalous Quartic Gauge Couplings modeled via **EFT dimension-8 operators** drastically enhance the cross-section at high energy scales ($\hat{s}$).

2D Fit Strategy

- **Observables:**
  BDT score (EW vs QCD) & m_T^{WZ}.
- **Binning:**
  4 BDT $\times$ 5 $m_T^{WZ} \rightarrow$ **1D histogram** (20 bins).
- **Test Statistic:**
  Profiled Likelihood Ratio $q(\vec{c})$ for **95% CL limits** on $f_j^{(8)}/\Lambda^4$.

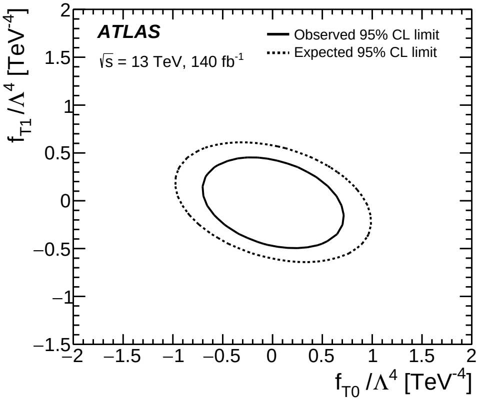


EFT Limits (Without Unitarization)

1D Limits (95% CL)

Coupling	Exp. [TeV^{-4}]	Obs. [TeV^{-4}]
f_{T0}/Λ^4	$[-0.80, 0.80]$	$[-0.57, 0.56]$
f_{T1}/Λ^4	$[-0.52, 0.49]$	$[-0.39, 0.35]$
f_{T2}/Λ^4	$[-1.6, 1.4]$	$[-1.2, 1.0]$
f_{M0}/Λ^4	$[-8.3, 8.3]$	$[-5.8, 5.6]$
f_{M1}/Λ^4	$[-12.3, 12.2]$	$[-8.6, 8.5]$
f_{M7}/Λ^4	$[-16.2, 16.2]$	$[-11.3, 11.3]$
f_{S02}/Λ^4	$[-14.2, 14.2]$	$[-10.4, 10.4]$
f_{S1}/Λ^4	$[-42, 41]$	$[-30, 30]$

2D Contour (f_{T0} vs f_{T1})



The EFT is not a complete model. Non-zero dimension-8 operators violate tree-level unitarity at high energies.

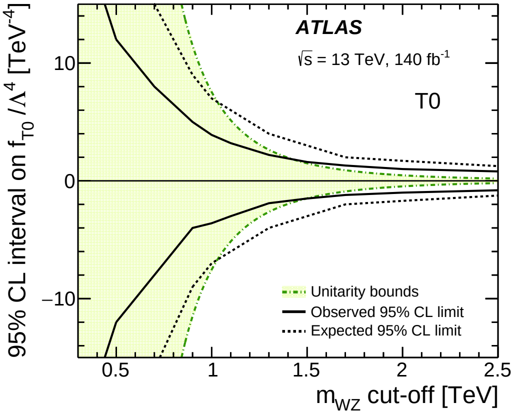
The "Clipping" Technique

- EFT contributions are removed above a cut-off scale E_c ($m_{WZ} > E_c$).
- Replaced by SM predictions to restore unitarity.

Unitarized Limits (at crossing bound):

- $f_{T0}/\Lambda^4 \in [-1.5, 1.6]$ TeV^{-4}
- $f_{T1}/\Lambda^4 \in [-0.7, 0.6]$ TeV^{-4}

Evolution of 95% CL limits vs cut-off scale $\Rightarrow$



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Conclusions: VBS as a Precision Probe for the EWK Sector

From Discovery to Precision Measurement (2015-2018, 140 fb^{-1})

- **Integrated cross-section:** $\sigma_{WZjj\text{-EW}} = 0.368 \pm 0.037$ (stat) $\pm\, 0.059$ (syst) fb.
- **Differential measurements** (via Bayesian Unfolding) on aQGC-sensitive variables.

Standard Model Verification and Modelling Challenges

- **EW Production:** Excellent Data/Theory agreement in the Signal Region.
- **QCD Background:** Tension of $\mathbf{1.8}\sigma$ (Data $\sim 30\%$ lower than nominal MadGraph).

Limits on New Physics (EFT) and Unitarity

- **95% CL limits** set on 9 (8) dimension-8 operators without (with) consideration of tree-level unitarity violation.
- These constraints are similar to those obtained bay the CMS Collaboration.

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Thank you for listening!

Any questions?

BACKUP

How to eliminate leptons ambiguity?

- Leptons and final-state neutrinos that do not originate from hadron or τ-lepton decays, are matched to the $W^{\pm}$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m^2_{(\ell^+,\ell^-)} - \left(m_Z^{\mathrm{PDG}}\right)^2 + \mathrm{i}\ \Gamma_Z^{\mathrm{PDG}}\ m_Z^{\mathrm{PDG}}} \right|^2 \times \left| \frac{1}{m^2_{(\ell',\nu_{\ell'})} - \left(m_W^{\mathrm{PDG}}\right)^2 + \mathrm{i}\ \Gamma_W^{\mathrm{PDG}}\ m_W^{\mathrm{PDG}}} \right|^2$$

Monte Carlo Simulation of the Signal

The simulation of $W^{\pm}Zjj$ events was primarily generated using the `MadGraph5_aMC@NLO2.6.5` MC event generator, interfaced with `Pythia8.240`. The simulation was divided into distinct components:

- **Electroweak production** $W^{\pm}Zjj-$ **EW**: The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α^6_{EW}).
- **QCD production** $W^{\pm}Zjj-$ **QCD**: Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference** $W^{\pm}Zjj$**-INT**: Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.

In addition to the $W^{\pm}Zjj-$ QCD production, there are other processes that can mimic the signal in the detector.

- **Backgrounds with *prompt* (real) leptons:** Processes such as the production of ZZ pairs, WW, tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Estimated directly using data-driven techniques.

All generated MC events were passed throught the ATLAS detector simulation, based om `Geant4`, and processed using the same recostruction software used for the data.

Expected and Observed Numbers of Events Pre-Fit

	SR ($N_{\text{jet}} = 2$)	SR ($N_{\text{jet}} \geq 3$)	b-CR	ZZ-CR
Data	169	477	666	210
Total pred.	231 ± 12	550 ± 50	660 ± 40	205 ± 11
$W^{\pm}Zjj-$ EW	65.0 ± 3.5	60 ± 6	4.82 ± 0.28	0.725 ± 0.014
$W^{\pm}Zjj-$ QCD	125 ± 9	380 ± 50	77 ± 18	6.2 ± 0.7
$W^{\pm}Zjj$-INT	1.3 ± 0.6	5.3 ± 2.6	0.58 ± 0.29	0.22 ± 0.11
$t\bar{t} + V$	0.66 ± 0.04	20.2 ± 0.7	289 ± 10	9.89 ± 0.28
tZj	8.78 ± 0.34	19.7 ± 1.2	134 ± 4	0.432 ± 0.005
ZZ-QCD	9.6 ± 0.4	32.0 ± 2.5	10.1 ± 0.6	159 ± 9
ZZ-EW	2.2 ± 0.6	4.4 ± 1.1	0.25 ± 0.06	23 ± 6
VVV	0.41 ± 0.10	2.0 ± 0.5	0.39 ± 0.10	4.1 ± 1.1
Misid. leptons	18 ± 4	28 ± 7	150 ± 40	1.7 ± 0.5

The Global Likelihood Fit

Simultaneous Extended Binned Negative Log-Likelihood:

$$-\ln \mathcal{L}(\boldsymbol{\mu}, \boldsymbol{K}, \boldsymbol{\nu}) = \sum_r \sum_i \left(\mu_{\text{exp}}^{[r,i]} - N^{[r,i]} \ln \mu_{\text{exp}}^{[r,i]} \right) + \sum_j \frac{\nu_j^2}{2}$$

- **Regions (r):** SR ($N_{\text{jets}} = 2$), SR ($N_{\text{jets}} \geq 3$), b-CR, ZZ-CR.
- **Observables ($N^{[r,i]}$):** bins BDT score in SRs and in b-CR and m_{jj} in ZZ-CR.
- **Parameters of Interest ($\boldsymbol{\mu}$):** Strength modifiers μ_{EW} and μ_{QCD}.
- **Unconstrained Parameters ($\boldsymbol{K}$):** Background normalizations ($K_{t\bar{t}V}, K_{tZj}, K_{ZZ}$) extracted dynamically from CRs.
- **Nuisance Parameters ($\boldsymbol{\nu}$):** Theoretical/experimental systematics, constrained by the Gaussian penalty term $\nu_j^2/2$.